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MILITARY STANDARD

FASTENER TEST METHODS

METHOD 22,

RECEPTACLE PUSH-OUT,

PANEL FASTENERS

FSC 53GP

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 22, Receptacle Push-Out, Panel Fasteners

MIL-STD-1312-22

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

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FOREWORD

This standard sets forth a standard test procedure for determining the push-out capabilities receptacle assembly of structural panel fasteners.

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1. SCOPE

1.1 Applicability. This test method covers the procedure and apparatus required for testing the push-out capability of receptacle assemblies of structural panel fasteners.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MS20426	Rivet, Solid, Countersunk 100 Degrees, Precision Head, Aluminum and Aluminum Alloy

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. DEFINITIONS

Not applicable

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Push-out test device. The push-out test device shall be capable of applying an axial force to the stud member of the receptacle assembly.

4.1.1.1 Force gage. A force gage capable of indicating the applied test load within an accuracy of ±1% may be utilized. A typical arrangement is shown in figure 1.

4.1.1.2 Dead weight loading. In lieu of the force gage specified in 4.1.1.1, the "Dead Weight" loading method may be utilized to apply an axial force to the receptacle assembly.

4.1.1.3 Deflection. The axial deflection of the receptacle assembly with respect to the attached plate shall be measured with a dial indicator which is accurate to within 0.0005 inch.

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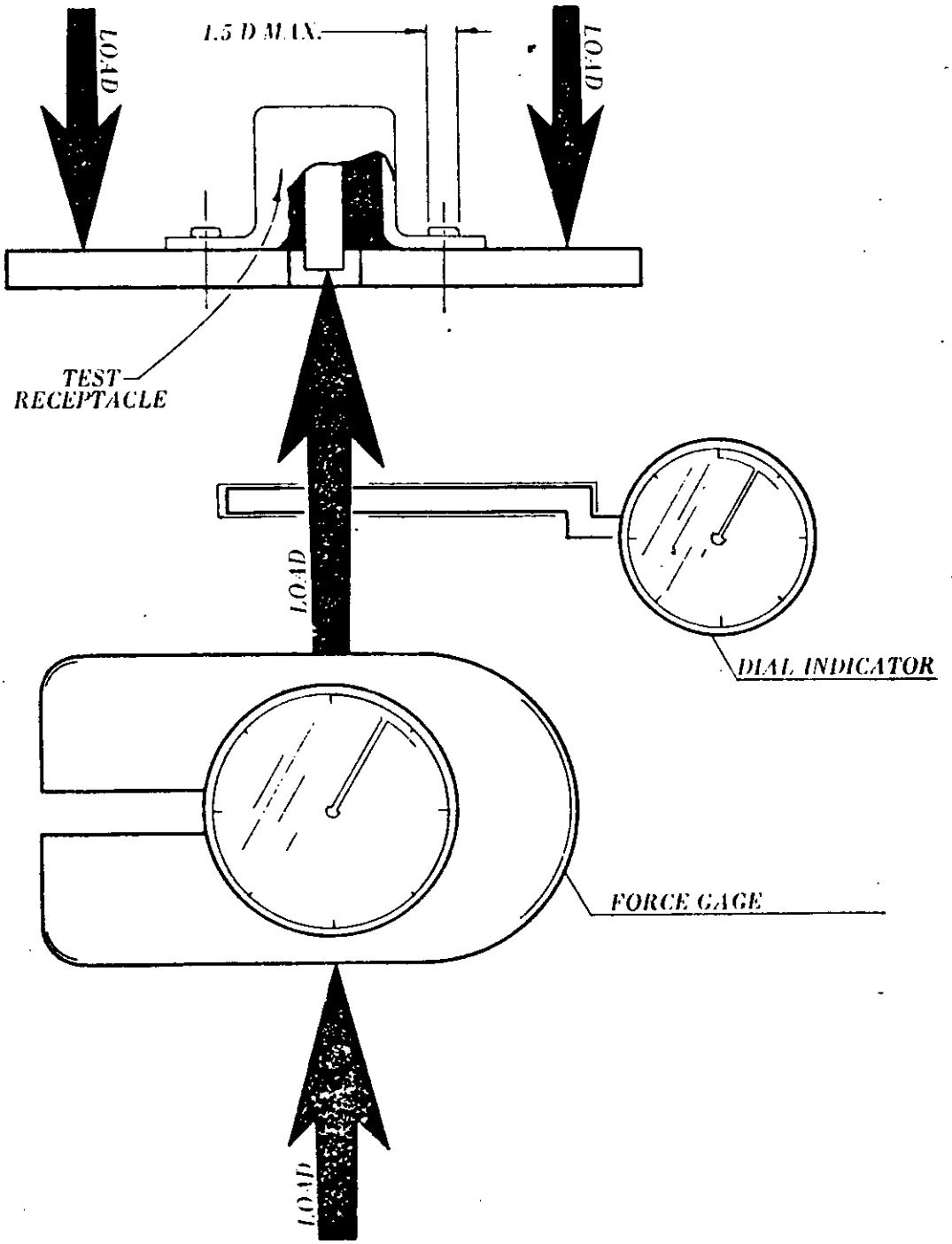


FIG. 1

FIGURE 1. Typical push-out fixture.

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5. DETAIL REQUIREMENTS

5.1 Test procedure.

5.1.1 Push-out test. Unless otherwise specified, securely attach receptacle assembly to a rigid test plate by means of MS20426 AD rivets. Apply push-out test load axially to the receptacle assembly through a simulated stud of the same type fastener material being tested and sufficiently long to transmit the entire test load to the receptacle components. The test load shall be applied to the head of the simulated stud until the requirement is reached. The load shall be removed and the permanent set measurement recorded. Failure of the attaching rivets is not to be interpreted as a failure of the receptacle.

6. NOTES

6.1 Test report. The test report shall include the following data:

a. Fastener Description.

(1) Part number.

(2) Lot identification.

(3) Material.

(4) Heat treat.

(5) Grip length.

(6) Mating part.

(7) Measured fastener diameter.

b. Test Machine.

(1) Model and serial number.

(2) Calibration date.

c. Extensometer and gage length.

d. Ultimate load.

e. Test method (where options are available).

f. Test load and load rate.

g. Test duration.

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h. Type of failure.

i. Results of all inspection.

j. Specimen preparation.

k. Other as specified.

l. Date of test.

m. Test performed by.

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Navy - AS
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Navy - AS, SH
Air Force - 11
DLA - IS

Preparing activity:
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